P1 = KN,

where:

N = the number of passengers for which the ship is to be certified; and

K = 0.056L

However, where the value of KN is greater than the sum of P and the whole volume of the actual passenger spaces above the bulkhead deck, the figure to be taken as P1 is that sum or two-thirds KN, whichever is the greater.

3.3 Where practicable, the power bilge pumps shall be placed in separate watertight compartments and so arranged or situated th at these compartments will not be flooded by the same damage. If the main propulsion m achinery, auxiliary machinery and boilers are in two or more watertight compartments, th e pumps available for bilge service shall be

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SOURCE: MSC.194(80).pdf
the same damage. If the main propulsion m achinery, auxiliary machinery and boilers are in two or more watertight compartments, th e pumps available for bilge service shall be distributed as far as is possible throughout these compartments.

3.4 On a ship of 91.5 m in length and upwards or having a bilge pump numeral, calculated in accordance with paragraph 3.2, of 30 or more, the arrangements shall be such that at least one power bilge pump sh all be available for use in all flooding conditions which the ship is required to withstand, as follows:
RESOLUTION MSC.194(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED-
50 -

.1 one of the required bilge pumps sha ll be an emergency pump of a reliable submersible type having a source of power situated above the bulkhead deck; or

.2 the bilge pumps and their sources of power shall be so distributed throughout the length of the ship that at least one pump in an undamaged compartment will be available.

3.5 With the exception of additional pu mps which may be provided for peak

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SOURCE: MSC.194(80).pdf
throughout the length of the ship that at least one pump in an undamaged compartment will be available.

3.5 With the exception of additional pu mps which may be provided for peak compartments only, each required bilge pump sha ll be so arranged as to draw water from

any space required to be drained by paragraph 2.1.

3.6 Each power bilge pump sh all be capable of pumping water through th e required main bilge pipe at a speed of not less th an 2 m/s. Independe nt power bilge pumps situated in machinery spaces shall have direct suctions from these spaces, except that not more than two such suctions shall be required in any one space. Where two or more such suctions are provided, there shall be at least one on each side of the ship. The Administration may require independent power bilge pumps situated in other spaces to have separate direct suctions. Direct sucti ons shall be suitably arranged and those in a machinery space shall be of a diameter not less than that required for the bilge main.

3.7.1 In addition to the direct bilge sucti on or suctions required by paragraph 3.6,

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SOURCE: MSC.194(80).pdf
machinery space shall be of a diameter not less than that required for the bilge main.

3.7.1 In addition to the direct bilge sucti on or suctions required by paragraph 3.6,
a direct suction from the main circulating pump leading to the drainage level of the machinery space and fitted with a non-return valve shall be provided in the machinery space. The diameter of this direct suction pipe shall be at least two thirds of the diameter of the pump inlet in the case of steamships, a nd of the same diameter as the pump inlet in the case of motorships.

3.7.2 Where in the opinion of the Administ ration the main circulating pump is not suitable for this purpose, a direct emergency bilge suction shall be led from the largest available independent power driven pump to the drainage level of the machinery space; the suction shall be of the same diameter as the main inlet of the pump used. The capacity of the pump so connected shall exc eed that of a required bilge pump by an amount deemed satisfactory by the Administration.

3.7.3 The spindles of the sea inlet and direct suction valves shall extend well above the engine-room platform.

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SOURCE: MSC.194(80).pdf
amount deemed satisfactory by the Administration.

3.7.3 The spindles of the sea inlet and direct suction valves shall extend well above the engine-room platform.

3.8 All bilge suction piping up to the connection to the pumps shall be independent of other piping.

3.9 The diameter d of the bilge main shall be calculated according to the following formula. However, the actual internal diameter of the bilge main may be rounded off to the nearest standard size acceptable to the Administration:

) (1.68 25D B Ld ++ =
where:
d is the internal diameter of the bilge main (millimetres);

L and B are the length and the breadth of the ship (metres) as defined in regulation 2; and
RESOLUTION MSC.194(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE INTERNATIONAL CONVENTION FOR THE SAFETY OF LIFE AT SEA, 1974, AS AMENDED-51 -

D is the moulded depth of the ship to th e bulkhead deck (metres) provided that, in a ship having an enclosed cargo space on the bulkhead deck which is internally drained in accordance with the requirements of paragraph 2.6.2 and which extends for the full length of the ship, D shall be measured to the next deck above the

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SOURCE: MSC.194(80).pdf
a ship having an enclosed cargo space on the bulkhead deck which is internally drained in accordance with the requirements of paragraph 2.6.2 and which extends for the full length of the ship, D shall be measured to the next deck above the bulkhead deck. Where the enclosed cargo spaces cover a lesser length, D shall be taken as the moulded depth to the bulkhead deck plus lh /L where l and h are the aggregate length and height respectively of the enclosed cargo spaces (metres).
The diameter of the bilge branch pipes shall meet the requirements of the Administration.

3.10 Provision shall be made to prevent the compartment served by any bilge suction pipe being flooded in the event of the pi pe being severed or otherwise damaged by collision or grounding in any other compartment. For this purpose, where the pipe is at any part situated nearer the side of the ship than one fifth of the breadth of the ship (as defined in regulation 2 and measured at right angles to the centreline at the level of the deepest subdivision load line), or is in a duct keel, a non-return valve shall be fitted to the pipe in the compartment containing the open end.

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SOURCE: MSC.194(80).pdf
(as defined in regulation 2 and measured at right angles to the centreline at the level of the deepest subdivision load line), or is in a duct keel, a non-return valve shall be fitted to the pipe in the compartment containing the open end.

3.11 Distribution boxes, cocks and valves in connection with the bilge pumping system shall be so arranged that, in the event of flooding, one of the bilge pumps may be operative on any compartment; in addition, damage to a pump or its pipe connecting to the bilge main outboard of a line drawn at one fifth of the breadth of the ship shall not put the bilge system out of action. If there is only one system of pipes common to all the pumps, the necessary valves fo r controlling the bilge suctions must be capable of being operated from above the bulkhead deck. Where in addition to the main bilge pumping system an emergency bilge pumping system is provided, it shall be independent of the main system and so arranged that a pump is capable of operating on any compartment under flooding condition as specified in paragr aph 3.1; in that case only the valves

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SOURCE: MSC.194(80).pdf

main system and so arranged that a pump is capable of operating on any compartment under flooding condition as specified in paragr aph 3.1; in that case only the valves necessary for the operati on of the emergency system need be capable of being operated from above the bulkhead deck.

3.12 All cocks and valves referred to in paragraph 3.11 which can be operated from above the bulkhead deck shall have their controls at their place of operation clearly marked and shall be provided with means to indicate whether they are open or closed.

4 Cargo ships

At least two power pumps connected to the ma in bilge system shall be provided, one of which may be driven by the propulsion machinery. If the Administratio n is satisfied that the safety of the ship is not impaired, bilge pumping arrangements may be dispensed with in particular compartments."

CHAPTER II-2
CONSTRUCTION FIRE PROTECTION, FIRE DETECTION AND FIRE EXTINCTION

Regulation 4 Probability of ignition

3 In paragraph 5.2.4, the reference to "regul ation II-1/25-9.2" is replaced by the reference to "regulation II-1/13-1.2".